

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: French Pharmaceutical Regulatory Affairs — Document Compliance NLP

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Data sources span scientific literature, clinical trials, clinical cases, and speech transcriptions [Q19], but the regulatory-document content footprint is small and bounded. QUAERO's EMEA subset and MANTRAGSC's fr_emea provide the only authentic drug-leaflet content (QUAERO-D2, QUAERO-D5, MANTRAGSC-D1, MANTRAGSC-D2), and these are used only for NER. Dataset analysis confirms substantial off-domain content dilution: ~30–40% of DEFT2020 is encyclopedic (railway, beekeeping, biography — DEFT2020-D14, DEFT2020-D15, DEFT2020-D16), QUAERO/medline includes historical and biographical titles (QUAERO-D16, QUAERO-D17, QUAERO-D18), MORFITT contains substantial veterinary and non-French-population content (MORFITT-D9, MORFITT-D8, MORFITT-D31), and DiaMED is sourced entirely from Sub-Saharan/North African clinical cases (DIAMED-D9, DIAMED-D11). Regulatory-specific vocabulary (INNs distinguished from brand names, ATC codes, excipient nomenclature, MedDRA terms, EMA posology templates) is not a deliberate coverage target in any dataset. No French overseas territory or tropical pathology content appears as a deliberate target — only incidental references in QUAERO-D11 and MORFITT-D17. Web search confirmed absence of DOM-specific labeling content [WEB-1, WEB-3].

ID	Evidence
Q19	"A diverse range of data origins: Scientific literature, clinical trials, clinical cases, speech transcriptions, and more as described in Table 2." (p.2)
MANTRAGSC-D1	"Chaque comprimé contient 500 mg de ranolazine." — Standard drug labeling INN+dosage sentence
QUAERO-D2	TYSABRI 300 mg solution à diluer pour perfusion natalizumab Chaque flacon de 15 ml de concentré contient 300 mg de natalizumab (20 mg / ml) — Brand name, INN, dosage, form, concentration co-occurring in regulatory sentence
MANTRAGSC-D2	"la posologie recommandée d' Agenerase solution buvable est de 17 mg (1,1 ml)/ kg trois fois par jour, sans excéder la dose maximale de 2800 mg par jour" — Posology statement with dose thresholds from drug leaflet
QUAERO-D5	Des épisodes de troubles psychiatriques aigus , tels que hallucinations , réactions paranoïdes , hostilité , délire , psychose et réactions maniaques ont été rapportés chez des patients traités par le ziconotide . — Safety warning listing adverse psychiatric events; drug name tagged CHEM
MORFITT-D8	La région du Nord-Ouest de l'Ontario présente un taux élevé et documenté d'infections de la peau et des tissus mous causées par une souche de Staphylococcus aureus méthycillino-résistante d'origine communautaire (SARM-C). — Canadian setting — geographic mismatch with metropolitan France regulatory context
MORFITT-D9	L'otite externe est une maladie multifactorielle fréquente chez le chien. — Canine ear disease — veterinary content irrelevant to human regulatory compliance
DIAMED-D9	Communication interventriculaire ischémique... dans le service de cardiologie du CHU-Yalgado Ouedraogo de Ouagadougou (Burkina Faso) — Explicitly Burkinabé hospital setting — not metropolitan France
DIAMED-D11	Ces douleurs étaient consécutives à une alimentation importante suite à la rupture du jeûn durant le mois de ramadan. — Ramadan context confirms African/North African origin; cultural/geographic mismatch
QUAERO-D11	Le paludisme chimiorésistant en France . — Drug-resistant malaria in France; incidental tropical disease relevance
DEFT2020-D14	Entre Perpignan et Villefranche, il subsiste de très nombreux poteaux caténaires datant des premiers essais en 12 KV 16 2/3 Hz... — Railway infrastructure; entirely off-domain
DEFT2020-D15	Boris Fiodorovitch Godounov, en russe : Борис Фёдорович Годунов (v.1551–Moscou, 13 avril 1605), gouverne la Russie... — Russian historical biography; off-domain
DEFT2020-D16	Certains apiculteurs sélectionnent leurs reines afin de favoriser au mieux la production. — Beekeeping content; off-domain

QUAERO-D16	L' apport des inventaires a la connaissance de la demographie parisienne ancienne : le regne de Francois Ier — Historical demography article title; entirely off-domain, all O tags
QUAERO-D17	De la médecine factuelle (evidence-based medicine) au ' libre arbitre ' . — Philosophical/methodological title; no biomedical entity, all O
MORFITT-D17	l'étude de l'épidémie de Chikungunya, un alphavirus transmis par Aedes aegypti et Aedes albopictus, survenue dans l'océan Indien en 2004-2007. — Chikungunya epidemic in Indian Ocean — relevant to overseas territory disease vocabulary
QUAERO-D18	LÉON GRIMBERT 1860 - 1931 . — Biographical citation; zero regulatory relevance, all O tags
MORFITT-D31	des médecins internes en formation en Arabie saoudite — Saudi Arabian medical intern study — non-metropolitan France context
WEB-1	French overseas territories are integral DOM under Article 73 — same regulatory framework applies (https://www.insee.fr/en/metadonnees/definition/c2316)
WEB-3	DrBenchmark paper makes no mention of French overseas territory vocabulary coverage (https://arxiv.org/abs/2402.13432)

Strengths

- QUAERO/emea and MANTRAGSC/fr_emea contain authentic EMA drug leaflet sentences with posology, contraindications, adverse events, and excipient lists (QUAERO-D1, QUAERO-D2, QUAERO-D5, MANTRAGSC-D1)
- Pharmaceutical vocabulary is collectively present across multiple datasets — drug names, dosages, administration routes, and pharmacovigilance-adjacent language (DEFT2021-D1, DEFT2021-D5, PXCORPUS-D1)
- All text is in standard written French biomedical register, consistent with deployment language requirements (CLISTER-D1, ESSAI-D1) [Q19]
- Specific contraindication phrasing for pregnancy/lactation, pediatric, and hypersensitivity populations present in EMA-style language (QUAERO-D6, QUAERO-D7, QUAERO-D8)

ID	Evidence
Q19	"A diverse range of data origins: Scientific literature, clinical trials, clinical cases, speech transcriptions, and more as described in Table 2." (p.2)
QUAERO-D1	Phosphate de sodium , monobasique , monohydraté Phosphate de sodium , dibasique , heptahydraté Chlorure de sodium Polysorbate 80 (E433) Eau pour préparation injectable — Excipient nomenclature list with E-numbers from drug leaflet
MANTRAGSC-D1	"Chaque comprimé contient 500 mg de ranolazine." — Standard drug labeling INN+dosage sentence
DEFT2021-D1	Mme J.K. est traitée pour une fibrillation auriculaire depuis deux ans avec le sulfate de quinidine 200 mg trois fois par jour... La prescription de clarithromycine nous semblerait, dans ce cas précis, d'un rapport bénéfice/risque très bas et potentiellement torsadogénique — Drug interaction case with benefit-risk reasoning; partial overlap with pharmacovigilance language
PXCORPUS-D1	primperan 10 milligrammes comprimés 1 comprimé en cas de nausée toutes les 8 heures pendant 14 jours — Brand drug with dose, form, indication, frequency, duration — deployment-relevant posology structure
CLISTER-D1	Une mastectomie était réalisée avec curage axillaire." / "Une mastectomie avec curage axillaire ont été réalisés. — Near-paraphrase in standard French clinical language
ESSAI-D1	Evaluer l' hypothèse selon laquelle une chimiothérapie à hautes doses avec une combinaison de Busulfan et de Melphalan (BU-MEL) permettra d' obtenir une survie sans événement à 3 ans supérieure à une consolidation par une association de Carboplatine VP16 et Melphalan chez des patients porteurs d' un neuroblastome à haut risque. — Formal French biomedical prose with drug names in clinical trial objective format
QUAERO-D2	TYSABRI 300 mg solution à diluer pour perfusion natalizumab Chaque flacon de 15 ml de concentré contient 300 mg de natalizumab (20 mg / ml) — Brand name, INN, dosage, form, concentration co-occurring in regulatory sentence
QUAERO-D5	Des épisodes de troubles psychiatriques aigus , tels que hallucinations , réactions paranoïdes , hostilité , délire , psychose et réactions maniaques ont été rapportés chez des patients traités par le ziconotide . — Safety warning listing adverse psychiatric events; drug name tagged CHEM
DEFT2021-D5	Hydrocortisone 300 mg IV immédiatement ; Traitement Hydroxyzine 25 mg par voie orale toutes les 6 heures — Posology-adjacent annotation of drug, dose, route, frequency
QUAERO-D6	Par conséquent , Refludan ne doit pas être administré à la femme enceinte ou qui allaite . — Standard contraindication phrasing for pregnancy/lactation in regulatory text
QUAERO-D7	Prialt ne doit pas être utilisé chez l' enfant . — Pediatric contraindication standard phrasing
QUAERO-D8	N' utilisez jamais TYSABRI ... Si vous êtes allergique (hydpersensible) au natalizumab ou à l' un des autres composants contenus dans TYSABRI — Allergy/hypersensitivity contraindication in patient-facing PIL language

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — deployment requires France-specific regulatory vocabulary (INNs, ATC, excipients, EMA QRD posology phrasing, ANSM Feuille de style elements, blue-box CIP codes, List I/II classification) [WEB-5, WEB-6]. The benchmark provides only partial proxies via QUAERO/emea and MANTRAGSC/fr_emea.

IC-2: Partial alignment: French biomedical register matches, but cultural/institutional content is mixed. DiaMED sources Sub-Saharan African clinical cases (DIAMED-D9 Burkina Faso, DIAMED-D11 Ramadan context); MORFITT includes Canadian, Jordanian, Saudi, and Egyptian content (MORFITT-D8, MORFITT-D13, MORFITT-D20, MORFITT-D31); DEFT2021 includes Canadian pharmacy clinical content (DEFT2021-D15, DEFT2021-D16) — geographic mismatch with metropolitan French regulatory context.

IC-3: INSUFFICIENT DOCUMENTATION on Western-specific knowledge framing per se; however, dataset analysis reveals non-French and non-metropolitan content (Italian E3C-D8, English E3C-D7, Basque E3C-D6, German MANTRAGSC-D13) that introduces construct-irrelevant variance for a metropolitan-French regulatory deployment.

IC-4: INSUFFICIENT DOCUMENTATION — the benchmark paper does not document recruitment of regulatory-affairs annotators or French-overseas-territory annotators. Would need an annotator demographic breakdown by region and professional role.

IC-5: Major content validity concerns: (1) regulatory-document content is limited to two NER subsets; (2) substantial off-domain dilution in DEFT2020, QUAERO/medline, MORFITT; (3) no French overseas territory or tropical pathology vocabulary as deliberate coverage [WEB-3]; (4) ICD-10 clinical mapping content (DiaMED) reflects deployment's stated narrower scope but does not address regulatory entity types.

ID	Evidence
Q19	"A diverse range of data origins: Scientific literature, clinical trials, clinical cases, speech transcriptions, and more as described in Table 2." (p.2)
QUAERO-D2	TYSABRI 300 mg solution à diluer pour perfusion natalizumab Chaque flacon de 15 ml de concentré contient 300 mg de natalizumab (20 mg / ml) — Brand name, INN, dosage, form, concentration co-occurring in regulatory sentence
MANTRAGSC-D2	"la posologie recommandée d' Agenerase solution buvable est de 17 mg (1,1 ml)/ kg trois fois par jour, sans excéder la dose maximale de 2800 mg par jour" — Posology statement with dose thresholds from drug leaflet
E3C-D6	Azken hilabeteetan lkernek kodeina + ibuprofenoa hartu ditu, baina ez du hobekuntza handirik nabaritu — Basque text; irrelevant to French deployment
E3C-D7	A chest radiograph showed right upper lobe consolidation with volume loss, right para-tracheal and left hilar adenopathy — English clinical text; irrelevant to French regulatory deployment
MORFITT-D8	La région du Nord-Ouest de l'Ontario présente un taux élevé et documenté d'infections de la peau et des tissus mous causées par une souche de Staphylococcus aureus méthycilline-résistante d'origine communautaire (SARM-C). — Canadian setting — geographic mismatch with metropolitan France regulatory context
E3C-D8	Trattata con nutrizione parenterale(NP), volume iniziale di 240 ml/die(166 kcal) — Italian clinical text; irrelevant to French regulatory deployment
DIAMED-D9	Communication interventriculaire ischémique... dans le service de cardiologie du CHU-Yalgado Ouedraogo de Ouagadougou (Burkina Faso) — Explicitly Burkinabé hospital setting — not metropolitan France

DIAMED-D11	Ces douleurs étaient consécutives à une alimentation importante suite à la rupture du jeûn durant le mois de ramadan. — Ramadan context confirms African/North African origin; cultural/geographic mismatch
QUAERO-D11	Le paludisme chimiorésistant en France . — Drug-resistant malaria in France; incidental tropical disease relevance
MORFITT-D13	En 2007 et 2008, 4 423 adultes de Calgary ont répondu à des entrevues au téléphone fixe portant sur l'activité physique — Calgary-based physical activity study — non-French, non-pharmaceutical content
MANTRAGSC-D13	“Die empfohlene Dosis für Agenerase Lösung beträgt 17 mg... Amprenavir/kg Körpergewicht” — German-language example; not relevant to French deployment
DEFT2020-D14	Entre Perpignan et Villefranche, il subsiste de très nombreux poteaux caténaux datant des premiers essais en 12 KV 16 2/3 Hz... — Railway infrastructure; entirely off-domain
DEFT2021-D15	le pharmacien hésite à lui remettre la prescription... Tableau I : Interactions médicamenteuses dans le cas clinique — Canadian pharmacy clinical case; register and institutional context differ from metropolitan French regulatory documentation
DEFT2021-D16	acétaminophène 300 mg par voie orale — Canadian drug name convention ('acétaminophène' vs French 'paracétamol'); cross-regional register variation
QUAERO-D16	L'apport des inventaires à la connaissance de la démographie parisienne ancienne : le règne de François Ier — Historical demography article title; entirely off-domain, all O tags
MORFITT-D17	l'étude de l'épidémie de Chikungunya, un alphavirus transmis par Aedes aegypti et Aedes albopictus, survenue dans l'océan Indien en 2004-2007. — Chikungunya epidemic in Indian Ocean — relevant to overseas territory disease vocabulary
MORFITT-D20	79 patients au total ont été recrutés à Amman (Jordanie) en 2015. — Jordanian oncology study — non-French geographic context
MORFITT-D31	des médecins internes en formation en Arabie saoudite — Saudi Arabian medical intern study — non-metropolitan France context
WEB-3	DrBenchmark paper makes no mention of French overseas territory vocabulary coverage (https://arxiv.org/abs/2402.13432)
WEB-5	ANSM-specific packaging requirements (CIP codes, Exploitant, List I/II) not reflected in benchmark content (https://www.pharmalex.com/thought-leadership/blogs/the-complex-landscape-of-french-regulatory-affairs/)
WEB-6	Variation types IA/IB/II directly map to threshold-based escalation requirements but are not evaluated in benchmark (https://alheha-consult.com/how-to-market-a-drug-in-france/)

Information Gaps

- Specific proportion of DEFT2020 that is regulatory-relevant vs. off-domain (sample-based estimate only)
- Whether QUAERO/emea full corpus contains higher proportion of high-value regulatory text than the sample suggests

Requires Expert Verification

- Whether the EMEA-leaflet content sampled in QUAERO and MANTRAGSC is representative of current ANSM-approved labeling phrasing

Remediation

Gap 1: No deliberate coverage of French overseas territory regulatory content or tropical pathology vocabulary (dengue, chikungunya, paludisme, Zika, leptospirose, leishmaniose) — identified as the highest secondary adaptation priority.

Recommendation: Construct a DOM-supplementation corpus of safety warnings and tropical pathology terminology, sourcing from ANSM circulars relevant to overseas territories and from clinical literature originating in DOM hospitals. Evaluate model degradation on this supplementary set before any DOM deployment.

Gap 2: Substantial off-domain dilution in DEFT2020 (~30–40% encyclopedic), QUAERO/medline (historical/biographical titles), and MORFITT (veterinary, non-French geographies) introduces construct-irrelevant variance for regulatory deployment.

Recommendation: When using DrBenchmark for fine-tuning or model selection, apply configuration-level filtering: prefer QUAERO/emea over QUAERO/medline, MANTRAGSC/fr_emea and fr_patents over fr_medline, exclude veterinary MORFITT abstracts, and weight DEFT2020 drug-leaflet pairs over encyclopedic pairs. Document filtering decisions for reproducibility.